

Grégoire Dhimoïla

✉ gregoire.dhimoila@ens-paris-saclay.fr

🔗 [Parabrele](#)

🎓 [Google Scholar](#)

Research Interests

Researcher in mechanistic interpretability and multimodal representation learning, with work on state space models and computational neuroscience.

Education

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| 2022 – 2026 | ENS Paris-Saclay
M2 MVA (2024 – 2025), M1 MPRI (2023 – 2024)
L3 Mathematics and Computer Science (2022 – 2023) — Ranked: 2/29 |
| 2020 – 2022 | Lycée Aux Lazaristes
Classe préparatoire MP* |

Research Experience

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| 2025 – 2026 | Brown University — Serre Lab
<i>Visiting Research Scholar</i> , advised by Thomas Serre. <ul style="list-style-type: none">• First author on two NeurIPS 2026 submissions and coauthor on one additional submission on multimodal representations and state space models [1,2,3]• Studied links between mechanistic interpretability and computational neuroscience• Developed frameworks for analyses of representation learning in multimodal models• Derived spectral formulations enabling parallel recurrent vision architectures. |
| Summer 2025 | DEEL / IRT Saint Exupéry
<i>Research Intern</i> , advised by Thomas Fel and Agustin Picard. <ul style="list-style-type: none">• First-author paper accepted at ICLR 2026 [4].• Introduced sparse decomposition methods separating shared and modality-specific embedding geometry• Enabled applications like semantic vector arithmetic and modality-gap-free embeddings. |
| 2024 – 2025 | Cadenza Labs
<i>Research Scholar</i> , advised by Walter Laurito. <ul style="list-style-type: none">• Co-first author paper accepted at EMNLP 2024 [5]• Worked on Contrast-Pair Clustering methods for CCS-style approaches and other mechanistic interpretability projects. |
| Summer 2024 | Goethe University Frankfurt
<i>Research Intern</i> , advised by Holger Dell. <ul style="list-style-type: none">• Developed computational graph structures based on aggregates of dictionary atoms• Investigated unsupervised behavioral control in neural networks through systematic interventions on the computational graph. |

Publications & Preprints

Conference Proceedings

- [4]** | **Grégoire Dhimoïla**, Thomas Fel, Victor Boutin, and Agustin Martin Picard
Cross-modal redundancy and the geometry of vision-language embeddings
ICLR 2026
- [5]** | Walter Laurito*, Sharan Maiya*, **Grégoire Dhimoïla***, Owen Ho Wan Yeung, Kaarel Hänni
Cluster-Norm for Unsupervised Probing of Knowledge
EMNLP 2024

Preprints / Under Review

- [1]** | **Grégoire Dhimoïla**, Victor Boutin, Agustin Martin Picard, Thomas Fel, and Thomas Serre
A Unifying Framework for Concept-Based Representational Similarity
NeurIPS 2026 submission
- [2]** | Drew Linsley*, **Grégoire Dhimoïla***, Alekh Karkada Ashok, Brian Kim, and Thomas Serre
Recurrent Vision in Parallel: Spectral Convolutional State Space Models
NeurIPS 2026 submission
- [3]** | Yu-Ang Cheng, Sixuan Chen, Zhouyang Lu, Xizheng Yu, **Grégoire Dhimoïla**, Thomas Serre
A Unified Spectral Theory of Multimodal Losses
NeurIPS 2026 submission